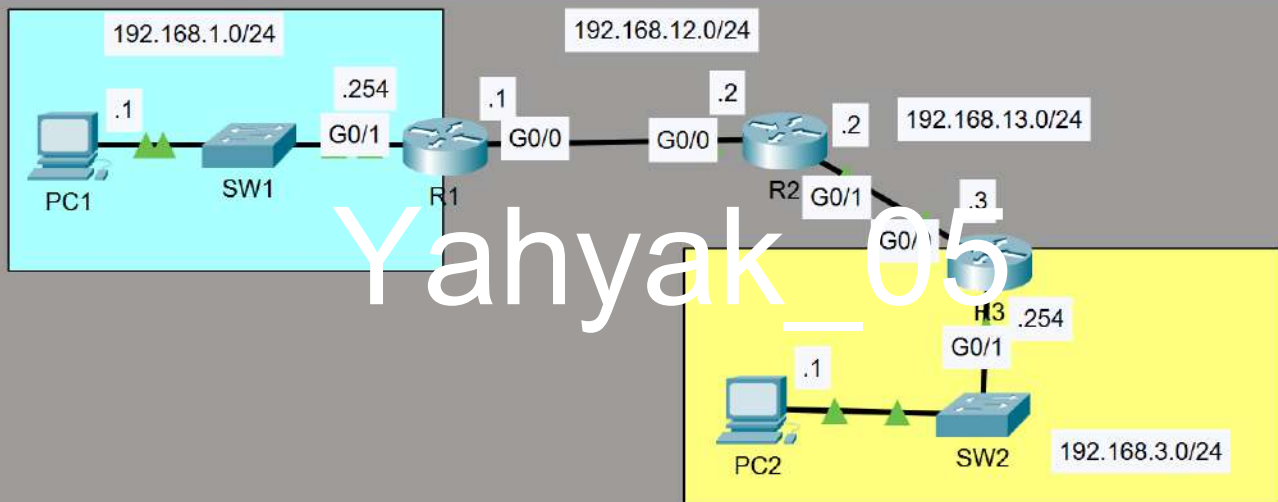


Troubleshooting Static IP routes



I began the troubleshooting lab by examining the network topology and identifying the routes that should exist between the different networks before testing connectivity

I verified the IP addressing configuration on the end devices to ensure the issue was not caused by incorrect host settings, I then attempted to Ping PC2 which failed

File PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.1
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.1.254
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-8D-08-1C-41-00-01-C7-
D7-50-66
DNS Servers.....: ::
                        0.0.0.0

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 0060.3EAE.6608
Link-local IPv6 Address.....:
IPv6 Address.....:
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-8D-08-1C-41-00-01-C7-
D7-50-66
DNS Servers.....: ::
                        0.0.0.0

C:\>
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



PC1



Physical

Config

Desktop

Programming

Attributes

Command Prompt



```
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-8D-08-1C-41-00-01-C7-
D7-50-66
DNS Servers.....: ::
                  0.0.0.0
```

```
C:\>
```

```
C:\>ping 192.168.3.1
```

```
Pinging 192.168.3.1 with 32 bytes of data:
```

```
Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

```
Ping statistics for 192.168.3.1:
```

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>ping 192.168.12.1
```

```
Pinging 192.168.12.1 with 32 bytes of data:
```

```
Reply from 192.168.12.1: bytes=32 time<1ms TTL=255
Reply from 192.168.12.1: bytes=32 time=26ms TTL=255
Reply from 192.168.12.1: bytes=32 time=1ms TTL=255
Reply from 192.168.12.1: bytes=32 time=1ms TTL=255
```

```
Ping statistics for 192.168.12.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 26ms, Average = 7ms
```

Yahyak_05

R1

Physical Config CLI Attributes

```
Gateway of last resort is not set

  192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.254/32 is directly connected, GigabitEthernet0/1
S       192.168.3.0/24 [1/0] via 192.168.12.3
  192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, GigabitEthernet0/0
L       192.168.12.1/32 is directly connected, GigabitEthernet0/0

R1#sh
% Int
R1#sh
ip route 192.168.3.0 255.255.255.0 192.168.12.3
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip route 192.168.3.0 255.255.255.0 192.168.12.3
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2
R1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.254/32 is directly connected, GigabitEthernet0/1
S       192.168.3.0/24 [1/0] via 192.168.12.2
  192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, GigabitEthernet0/0
L       192.168.12.1/32 is directly connected, GigabitEthernet0/0

R1(config)#
```

Copy Paste

here I identified that the Next-Hop between R1 and R2 was configured incorrectly. I deleted this static Route and configured the correct one

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R2

Physical Config CLI Attributes

249856K Bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2>

R2>show ipint brief

Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0	192.168.12.2	YES	manual	up
GigabitEthernet0/1	192.168.13.2	YES	manual	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down

R2>show ip route

Codes: L - local, C - connected, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

P - periodic downloaded static route

Gateway of last resort is not set

S 192.168.1.0/24 [1/0] via 192.168.12.1

S 192.168.3.0/24 is directly connected, GigabitEthernet0/0

192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.12.0/24 is directly connected, GigabitEthernet0/0

L 192.168.12.2/32 is directly connected, GigabitEthernet0/0

192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.13.0/24 is directly connected, GigabitEthernet0/1

L 192.168.13.2/32 is directly connected, GigabitEthernet0/1

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here on R2 the Next-Hop was being sent to the wrong interface



R2

Physical

Config

CLI

Attributes

```
GigabitEthernet0/0    192.168.12.2    YES manual up          up
GigabitEthernet0/1    192.168.13.2    YES manual up          up
GigabitEthernet0/2    unassigned      YES unset  administratively down down
Vlan1                 unassigned      YES unset  administratively down down
```

```
R2>show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
```

```
area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
S    192.168.1.0/24 [1/0] via 192.168.12.1
```

```
S    192.168.3.0/24 is directly connected, GigabitEthernet0/0
```

```
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C    192.168.12.0/24 is directly connected, GigabitEthernet0/0
```

```
L    192.168.12.2/32 is directly connected, GigabitEthernet0/0
```

```
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C    192.168.13.0/24 is directly connected, GigabitEthernet0/1
```

```
L    192.168.13.2/32 is directly connected, GigabitEthernet0/1
```

I deleted the previous configuration and
set the next hop to the correct interface

```
R2>enable
```

```
R2#show running-config | ip route
```

```
% Invalid input detected at '^' marker.
```

```
R2#show running-config | include ip route
```

```
ip route 192.168.1.0 255.255.255.0 192.168.12.1
```

```
ip route 192.168.3.0 255.255.255.0 GigabitEthernet0/0
```

```
R2#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R2(config)#ip route 192.168.3.0 255.255.255.0 GigabitEthernet0/1
```

```
%Default route without gateway, if not a point-to-point interface, may impact  
performance
```

```
R2(config)#
```

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R3

Physical Config CLI Attributes

P - periodic downloaded static route

Gateway of last resort is not set

192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.3.0/24 is directly connected, GigabitEthernet0/1

L 192.168.3.254/32 is directly connected, GigabitEthernet0/1

C 192.168.23.0/24 is directly connected, GigabitEthernet0/0

L 192.168.23.3/32 is directly connected, GigabitEthernet0/0

R3#sh run | include ip route

ip route 192.168.1.0 255.255.255.0 192.168.13.2

R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, i - IS-IS inter

area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.3.0/24 is directly connected, GigabitEthernet0/1

L 192.168.3.254/32 is directly connected, GigabitEthernet0/1

192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.23.0/24 is directly connected, GigabitEthernet0/0

L 192.168.23.3/32 is directly connected, GigabitEthernet0/0

R3#show ip int brief

Interface	IP-Address	OK?	Method	Status
GigabitEthernet0/0	192.168.23.3	YES	manual	up
GigabitEthernet0/1	192.168.3.254	YES	manual	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down

R3#

for R3, I discovered that the Next-Hop
was correct however the interface IP
address was misconfigured

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Source Destination

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US

Yahyak_05

Physical Config CLI Attributes

```
L    192.168.3.254/32 is directly connected, GigabitEthernet0/1
    192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.23.0/24 is directly connected, GigabitEthernet0/0
L    192.168.23.3/32 is directly connected, GigabitEthernet0/0

R3#show ip int brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0       192.168.23.3    YES manual up
GigabitEthernet0/1       192.168.3.254  YES manual up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface g0/0
R3(config-if)#ip add 192.168.13.3 255.255.255.0
R3(config-if)#exit
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.1.0/24 [1/0] via 192.168.13.2
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, GigabitEthernet0/1
L    192.168.3.254/32 is directly connected, GigabitEthernet0/1
    192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, GigabitEthernet0/0
L    192.168.13.3/32 is directly connected, GigabitEthernet0/0

R3#
```

Copy

Paste

Top



PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

X

```
Cisco Packet Tracer PC1>  
C:\>  
C:\>  
C:\>ping 192.168.3.1
```

After troubleshooting the routers and properly configuring the interfaces, I reattempted to ping PC2 from PC1

```
Pinging 192.168.3.1 with 32 bytes of data:
```

```
Request timed out.  
Request timed out.  
Request timed out.  
Reply from 192.168.3.1: bytes=32 time<1ms TTL=125
```

```
Ping statistics for 192.168.3.1:  
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.3.1
```

Yahyak_05

```
Pinging 192.168.3.1 with 32 bytes of data:
```

```
Reply from 192.168.3.1: bytes=32 time<1ms TTL=125  
Reply from 192.168.3.1: bytes=32 time=2ms TTL=125  
Reply from 192.168.3.1: bytes=32 time<1ms TTL=125  
Reply from 192.168.3.1: bytes=32 time<1ms TTL=125
```

```
Ping statistics for 192.168.3.1:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

```
C:\>|
```

SUCCESS!!